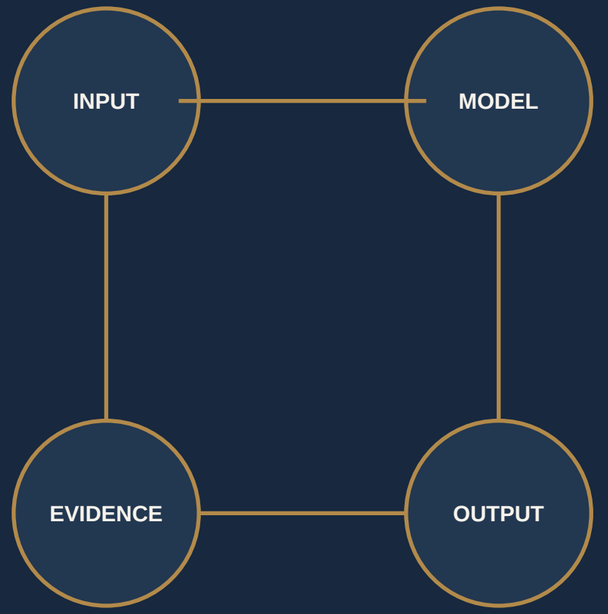
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

AI for Robotics

AI.208 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define how machine learning.
- Apply key methods including computer vision.
- Build or analyze a practical system involving planning.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

How machine learning

02

METHODS

Computer vision

03

SYSTEMS

Planning

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · How machine learning

How machine learning

02 · Computer vision

Computer vision

03 · Planning

Planning

04 · Reinforcement learning and int

Reinforcement learning and intelligent decision-making are integrated into robotic systems

05 · Mandatory practical: Build ar

Mandatory practical: Build an intelligent robotic application combining perception

06 · Learning and action

Learning and action

Core ideas, made explicit

AI.208 • Distinguish definitions, assumptions and evidence

- How machine learning
- Computer vision
- Planning
- Reinforcement learning and intelligent decision-making are integrated into robotic systems
- Mandatory practical: Build an intelligent robotic application combining perception

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.208 • Distinguish definitions, assumptions and evidence

- Learning and action

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving how machine learning.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using how machine learning.
- Implement or simulate a baseline using computer vision.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.